
IAL Physics Unit 2 (WPH12): A* Revision Guide

Waves and Electricity

Compiled from 21 WPH12 papers and their mark schemes (June 2019 – June 2026). Mark-scheme wording is quoted where the exact phrase earns the mark.

1. The paper and where the marks are

Format. 1 hour 30 minutes, 80 marks. Section A: 10 multiple-choice questions (1 mark each). Section B: about 8–10 structured questions (70 marks), always including one asterisked 6-mark extended-response question. Roughly half the marks are electricity and half are waves and particle nature; nearly every question ends with “is the student correct?”.

Topic	How often (21 papers)	What is actually asked
Photoelectric effect, photons, energy levels	Every paper	$E = hf$ with $\text{eV} \leftrightarrow \text{J}$; $hf = \Phi + \frac{1}{2}mv_{\text{max}}^2$; why the wave model fails (6-marker); line spectra; number of photons per second.
Potential dividers, thermistors, LDRs	Every paper	Ratio of p.d.s = ratio of resistances; read R from a graph; “what happens to the meter readings?”; sensor circuit 6-marker.
e.m.f. and internal resistance	Almost every paper	$\varepsilon = V + Ir$; V - I graph (intercept ε , gradient $-r$); circuit to measure them; lost volts.
Resistivity and drift velocity	Every paper	$R = \rho l/A$ (often a strip: $A = \text{width} \times \text{thickness}$); $I = nqvA$; resistivity method; metals vs semiconductors.
Refraction, critical angle, TIR	Every paper	Snell’s law with angles measured from a diagram; $n = c/v$; does TIR occur?; optical fibres.
Stationary waves	Every paper	Formation (reflection + superposition); $v = \sqrt{T/\mu}$ with $v = f\lambda$; $\lambda = 2L$ (string) or $4L$ (closed tube); phase of points.
Diffraction grating and interference	Almost every paper	$n\lambda = d \sin \theta$ with $\tan \theta$ from distances; lines per mm; maximum order; path difference and phase.
Polarisation, ultrasound, intensity, de Broglie	Most papers	Definitions; filter behaviour; pulse-echo distance (half the time); $I = P/A$; $\lambda = h/p$ and electron diffraction.

2. How A* scripts collect marks

RULES TAKEN FROM THE MARK SCHEMES

1. **“Show that”**: give one more significant figure than the printed value (4.43 eV, not 4.4 eV); a full reverse calculation scores at most 3 marks. If stuck, use the printed value in the next part.
2. **Units**: a missing or wrong unit loses the final mark (except in “show that” questions where the unit is given). Convert mm → m, nm → m, μs → s, minutes → s, kWh → J ($\times 3.6 \times 10^6$) and eV → J ($\times 1.60 \times 10^{-19}$) **before** substituting.
3. **Radius, not diameter**: $A = \pi r^2 = \pi d^2/4$. Using the diameter is the commonest lost mark in resistivity and drift-velocity questions.
4. **“Is the student correct?”**: the last mark needs your number, a comparison (“0.94 V > 0.90 V”) and a stated conclusion. A correct conclusion from your own earlier value still scores (ecf).
5. **Graphs**: line of best fit first; gradient from a large triangle (over half the line); read the intercept. Read several wavelengths or periods from a trace and divide.
6. **Words that are refused**: “atoms” for “electrons” in energy-level answers; “bending” for refraction (say “change of direction”); “out of phase” for destructive interference (say “antiphase” or “phase difference of π ”); “size” for amplitude; “light” for electromagnetic radiation where a general answer is needed.

6-MARK EXTENDED RESPONSE: HOW IT IS SCORED

Six indicative content (IC) points. Marks for content: 6 IC → 4, 5-4 IC → 3, 3-2 IC → 2, 1 IC → 1, plus up to 2 marks for linkage. **6/6 needs all six points as a connected chain** (cause → effect, joined with “so”, “because”, “therefore”).

Unit 2 six-markers come from a small bank: photoelectric effect vs wave model; sensor (thermistor/LDR) potential divider; ultrasound pulse-echo; line spectra; interference of two sources; stationary waves in a tube; filament lamp vs thermistor I - V . All are modelled in section 9.

3. Charge, current, p.d., power

$I = \Delta Q / \Delta t$	Number of electrons = $Q/e = It/e$. Current is the rate of flow of charge.
$V = W/Q$	p.d. is the energy transferred per unit charge (between two points). $1 \text{ V} = 1 \text{ J C}^{-1}$; $C = \text{J V}^{-1}$; base units of V are $\text{kg m}^2 \text{ s}^{-3} \text{ A}^{-1}$.
$P = VI = I^2R = V^2/R$; $W = VIt$	Choose the version with the quantity that stays constant: same V (parallel, fixed supply) → V^2/R ; same I (series) → I^2R . Doubling V on a fixed resistor gives 4× the power.
$R = V/I$	Resistance at a point on an I - V curve is V/I at that point, not the gradient. Zero current means infinite resistance.

- **e.m.f.:** “the energy supplied (work done) per unit charge (by the source)” or “the work done moving unit charge around the whole circuit”, or the terminal p.d. when no current flows.
- **Base quantities and units:** current (A) is a base quantity; charge ($C = A\ s$), volt and power are derived. Base units of power: $kg\ m^2\ s^{-3}$.

I-V characteristics (MCQ in almost every paper)

Ohmic resistor	Straight line through the origin: R constant (at constant temperature).
Filament lamp	I increases at a decreasing rate (S-shape, symmetric). As V increases, current increases, so the temperature rises, so the lattice ions vibrate more, so the electrons collide with them more often, so the resistance increases.
Thermistor (NTC)	I increases at an increasing rate. As temperature rises, more conduction electrons are released, which outweighs the extra lattice vibration, so the resistance decreases.
Diode / LED	No current in reverse and almost none until the threshold (about 0.7 V for a silicon diode, about 1.6–2 V for an LED); after that the resistance falls sharply. Below threshold the resistance is almost infinite.
LDR	More light means more photons absorbed, so more conduction electrons and lower resistance.

Measuring an *I-V* curve: component, ammeter and variable resistor (or potential divider) in series with the supply; voltmeter in parallel with the component; a potential divider lets V go to zero; for a diode/LED include a protective resistor and the component in forward bias.

4. Circuits

- **Series:** the same current everywhere (conservation of charge); p.d.s add to the e.m.f. (conservation of energy); $R_T = R_1 + R_2$.
- **Parallel:** the same p.d. across each branch; currents add at a junction; $1/R_T = 1/R_1 + 1/R_2$. The total is always less than the smallest branch.
- **Derivations (3 marks, asked several times):** series: $V = V_1 + V_2$, so $IR_T = IR_1 + IR_2$, and divide by I . Parallel: $I = I_1 + I_2$, so $V/R_T = V/R_1 + V/R_2$; V is the same in parallel, so divide by V . Name the conservation law each time.
- **Lamps in series vs parallel:** in parallel each gets the full supply p.d. (brighter, and one failing does not affect the other); in series they share it. A lamp's resistance changes with p.d., so you cannot assume $R_{\text{series}} = 4 \times R_{\text{parallel}}$ (Jun 2023).
- **Two different lamps in series** do not each get their rated p.d., because the p.d. divides in the ratio of their resistances (Oct 2021).
- **Solar cells (6-mark, Jun 2026):** series gives the same current and a larger total p.d.; parallel gives the same p.d. and a larger total current; $P = VI$ is the same either way; parallel keeps working if one cell fails.

Potential dividers and sensors

$V_{\text{out}} = V_{\text{in}} \times R_2 / (R_1 + R_2)$	Equivalent: $V_1/V_2 = R_1/R_2$. Or find $I = V/R_{\text{total}}$, then $V = IR$ for each part. Either earns full marks.
Uniform wire (potentiometer)	p.d. $\propto$ length of wire, so $V = V_{\text{total}} \times (x/L)$.

ROUTINE: “WHAT HAPPENS TO THE METER READINGS?” (2-4 MARKS, EVERY PAPER)

1. State what happens to the sensor’s resistance (thermistor: T up $\rightarrow R$ down; LDR: light up $\rightarrow R$ down).
2. Total resistance changes, so the current changes the opposite way (for a fixed supply).
3. The sensor takes a smaller (or larger) **share** of the p.d., so the p.d. across the fixed resistor changes the other way.
4. Conclude for each meter. Never contradict yourself (“lower V so lower I ” for the same component loses the mark).

EXAMPLE (OCT 2025). Thermistor $56 \, \Omega$ in series with $300 \, \Omega$ across $6.0 \, \text{V}$: $V_{\text{th}} = 6.0 \times 56/356 = 0.94 \, \text{V} > 0.90 \, \text{V}$ switching level, so the heater has not switched off at $65 \, ^\circ\text{C}$.

e.m.f. and internal resistance

$\varepsilon = I(R + r) = V + Ir$	Ir is the “lost volts”: energy per coulomb dissipated inside the cell, so the terminal p.d. is less than the e.m.f.
$V = -rI + \varepsilon$	Compare with $y = mx + c$: plot V against I ; intercept = ε , gradient = $-r$. Two identical cells in series double both.
$R = \varepsilon(1/I) - r$	Plot R against $1/I$: gradient = ε , y-intercept = $-r$ (Jan 2023).

- **Circuit to measure ε and r :** cell, ammeter and variable resistor in series; voltmeter across the cell terminals. Vary R , record pairs of V and I , plot V against I .
- **Voltmeter reads ε** only when there is no current (open circuit), because then there is no p.d. across r .
- **Increasing the external R** decreases I , so less p.d. across r , so the terminal p.d. rises. A larger load current (motor, hill, more lamps in parallel) means more lost volts and more heating in the battery.
- Power wasted in the cell = I^2r ; power to the load = I^2R ; total = εI .

5. Resistivity and drift velocity

$R = \rho l/A$	Resistivity: “the resistance of a sample of unit length and unit cross-sectional area”, or $\rho = RA/l$ with terms defined. Units $\Omega \, \text{m}$. For a strip, A = width $\times$ thickness.
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$I = nqvA$	n = number of charge carriers (conduction electrons) per m³ ; v = drift velocity, the mean velocity of the free electrons (about 10^{-4} m s ⁻¹ in copper).
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- **Drift velocity comparisons:** in series the current is the same, so $v \propto 1/(nA)$. A thinner section (or a material with fewer carriers) has a faster drift. Doubling the diameter quarters the drift ($4\times$ area).
- **Metal vs semiconductor (repeated 2-3 marks):** heating a metal increases the amplitude of lattice vibrations, so more frequent electron-ion collisions, so R increases (n barely changes). Heating a semiconductor (thermistor) releases more conduction electrons (n increases), so R decreases.
- **Stretching a wire at constant volume (Jun 2026):** doubling the length halves A , so $R = \rho l/A$ increases $4\times$; $R \propto l^2$.
- **Same p.d., halve the length:** R halves, so I doubles, so v doubles (Jan 2022).

RESISTIVITY EXPERIMENT (METHOD QUESTION IN MANY PAPERS)

1. Measure the diameter with a **micrometer** (or digital calliper) at several points and orientations; take the mean; $A = \pi d^2/4$.
2. Wire, ammeter and variable resistor in series with the supply; voltmeter across the length of wire being tested (or use an ohmmeter).
3. Measure the length l between the contacts with a metre rule; record V and I , so $R = V/I$; repeat for several lengths.
4. Plot R against l : gradient = ρ/A , so $\rho = \text{gradient} \times A$. (Or plot V against l at constant I : $\rho = \text{gradient} \times A/I$.)
5. Use a small current (or switch off between readings) so the wire does not heat up and change its resistance.

6. Waves: properties, stationary waves, interference

$v = f\lambda$, $f = 1/T$	From a displacement- distance graph read λ ; from a displacement- time graph read T . You need both to get v . Amplitude does not affect speed.
Phase difference = (path difference/ λ) $\times 2\pi$	$\lambda/8 \rightarrow \pi/4$; $3\lambda/8 \rightarrow 135^\circ$; $1.5\lambda \rightarrow \pi$ (antiphase); $n\lambda \rightarrow$ in phase.
$v = \sqrt{T/\mu}$	T = tension (often mg), μ = mass per unit length (kg m ⁻¹). Combine with $v = f\lambda$: $f = (1/2L)\sqrt{T/\mu}$, so $f \propto \sqrt{T}$, $f^2 \propto T$.
$I = P/A$, point source $A = 4\pi r^2$	Intensity is power per unit area (W m ⁻²). Inverse square: doubling d quarters I . Solar panel at an angle: $P = IA \cos \theta$.
Transverse	Oscillations perpendicular to the direction of energy transfer (wave travel). Examples: light and all EM waves, waves on strings, S-waves.

Longitudinal	Oscillations parallel to the direction of energy transfer. Sound, ultrasound, P-waves. At the centre of a compression or rarefaction the particle displacement is zero ; pressure and density are maximum at compressions and minimum at rarefactions.
Coherent	Same frequency and a constant phase difference (not necessarily in phase or equal amplitude). Two separate lamps are not coherent.
Diffraction	Spreading out of waves as they pass through a gap or around an obstacle; greatest when the gap $\approx \lambda$.
Huygens' construction	Each point on a wavefront acts as a source of secondary wavelets; these superpose and their envelope gives the new wavefront. A wavefront is a line on which all points are in phase.

Stationary (standing) waves

- **Formation (2-3 marks):** the wave reflects at the end (fixed end, pulley, closed end, metal plate); the incident and reflected waves (same frequency, travelling in opposite directions) **superpose/interfere**. Nodes form where the interference is destructive (zero amplitude); antinodes where it is constructive (maximum amplitude).
- **Spacing:** node to node = $\lambda/2$; node to antinode = $\lambda/4$. A string with 3 loops: $L = 3\lambda/2$.
- **Phase:** all points between two adjacent nodes are in phase (with different amplitudes); points in adjacent loops are in antiphase. No net energy transfer.
- **String fundamental:** node at each end, so $\lambda = 2L$. **Tube closed at one end:** node at the closed end, antinode at the open end, so $\lambda = 4L$. **Both ends open:** $\lambda = 2L$. The antinode is actually slightly beyond the open end (end correction), so the true λ is longer and the calculated speed is too low.
- **Raising the pitch:** shorten the string, increase the tension, or use a lighter string (smaller μ). For a recorder or tube, a longer air column gives a lower frequency.
- **Graph method:** f^2 against T is a straight line through the origin with gradient $1/(4L^2\mu)$; node spacing d against $1/f$ has gradient $v/2$.
- **Progressive vs stationary sound (Jun 2023):** same frequency; the progressive wave transfers energy and has the same amplitude everywhere; a string wave is transverse, sound is longitudinal.

EXAMPLE (JUN 2022). 0.300 kg hanging mass, $\mu = 2.27 \times 10^{-3} \text{ kg m}^{-1}$, 30 Hz: $T = 2.94 \text{ N}$, $v = \sqrt{(2.94/2.27 \times 10^{-3})} = 36.0 \text{ m s}^{-1}$, $\lambda = 1.20 \text{ m}$, so nodes are 0.60 m apart.

Interference of two sources

- **Maximum (loud, bright, constructive):** path difference = $n\lambda$, so the waves arrive in phase. **Minimum (quiet, dark, destructive):** path difference = $(n + \frac{1}{2})\lambda$, so they arrive in antiphase (phase difference π).
- **Measuring the speed of sound with two speakers:** find a minimum, use Pythagoras for each path length, path difference = $\lambda/2$, then $v = f\lambda$. Use a microphone and oscilloscope rather than ears (the amplitude can be judged more precisely). Reflections from walls blur the maxima and minima.
- **Speed of sound with an oscilloscope:** start with the two traces in phase; move the microphone until they are next in phase; distance moved = λ ; read T from the timebase

over several cycles; $v = \lambda/T$. Measuring over several wavelengths reduces the percentage uncertainty.

- **Thin-film coating (Oct 2023):** path difference = $2 \times$ thickness; a minimum needs $2t = \lambda/2$ in the coating; $\lambda_{\text{air}} = n \times \lambda_{\text{coating}}$.

Diffraction grating

$n\lambda = d \sin \theta$	$d = 1/(\text{lines per metre})$; 300 lines $\text{mm}^{-1} \rightarrow d = 3.33 \times 10^{-6}$ m. Find θ from $\tan \theta = (\text{distance from central maximum})/(\text{grating-to-screen distance})$. Double-slit equations score nothing here.
Maximum order	$\sin \theta \leq 1$, so $n_{\text{max}} = d/\lambda$ rounded down . If $\sin \theta$ comes out above 1, that order does not exist.

- Bright maxima: the path difference between waves from adjacent slits is a whole number of wavelengths, so they arrive in phase and interfere constructively.
- **White light:** white central maximum; spectra either side, violet closest to the centre (shortest λ , smallest θ) and red furthest.
- **Reducing uncertainty:** measure between the first-order maxima on **both** sides and halve; increase the grating-to-screen distance; use higher orders. The grating must be perpendicular to the beam and parallel to the screen. Laser safety: never look along the beam; avoid reflective surfaces.

EXAMPLE (JAN 2026). $\tan \theta = 0.78/4.0$, so $\theta = 11.0^\circ$; $d = 635 \times 10^{-9} / \sin 11.0^\circ = 3.32 \times 10^{-6}$ m, so 301 lines per mm, matching the 300 mm^{-1} label.

7. Refraction, polarisation, ultrasound

$n_1 \sin \theta_1 = n_2 \sin \theta_2$; $n = c/v$	Angles are measured from the normal . When measuring from a diagram, draw the normal, use a protractor, and quote the angles. $n_1 v_1 = n_2 v_2$.
$\sin C = 1/n$ (to air); $\sin C = n_2/n_1$ in general	TIR conditions: light travelling from a higher to a lower refractive index (more to less optically dense) and angle of incidence greater than the critical angle.

- **Refraction explained (2 marks):** the wave slows down entering a denser medium; one part of the wavefront meets the boundary first and slows first, so the direction changes. Along the normal the whole wavefront slows at once, so there is no change of direction. Frequency is unchanged; speed and wavelength decrease.
- **Does TIR happen?** Either compare i with C , or calculate $n_1 \sin \theta_1/n_2$: if it exceeds 1 there is no refracted ray. Draw TIR with the angle of reflection equal to the angle of incidence.
- **Dispersion:** n is larger for violet than red, so violet refracts more, and at a near-critical angle violet can totally internally reflect while red escapes.
- **Optical fibres:** rays at different angles travel different path lengths and take different times, so pulses spread out (longer duration). Cladding has a refractive index just

below the core's, so the critical angle increases (less TIR, more light lost) but the core is protected. A lower- n cladding gives a smaller critical angle and more TIR.

- **Fingerprint sensor (Oct 2023):** where skin touches the glass, light refracts into the skin (dark); where there is air (valleys), TIR reflects light (bright).
- **Refractive index from a refractometer/tank:** measure i and r (or use trigonometry on given distances), then apply Snell's law and compare with a table.

Polarisation (3 marks)	Unpolarised: oscillations in all (many) planes. Plane polarised: oscillations in one plane, which includes the direction of wave travel , or in one direction, perpendicular to the direction of travel. Only transverse waves can be polarised, so polarisation is evidence that light is transverse.
Testing with one filter	Rotate a single polarising filter: unpolarised light gives constant intensity; polarised light varies from maximum to minimum every 90° .
Crossed filters	Two filters at 90° transmit nothing. A third filter between them at an angle lets some light through. Stressed plastic rotates the plane of polarisation, so bright regions show high stress.
Reflected light	Light reflected from water or glass is partially (horizontally) polarised; a vertical polarising filter (sunglasses) absorbs it. Phone screens emit polarised light, so rotating the phone 90° behind polarising glasses makes the screen go dark. At Brewster's angle, $\tan \theta_B = n_2/n_1$.
Radio waves / aerials	Signal strength varies as the aerial is rotated, so the waves are polarised (the aerial acts as a polarising filter).

ULTRASOUND PULSE-ECHO (CALCULATION EVERY YEAR; 6-MARK METHOD JAN 2020 AND JAN 2026)

1. Pulses of ultrasound are sent into the material.
2. They are partially reflected at a boundary between different media (a change in density), e.g. a crack or the back wall.
3. The time between sending the pulse and receiving the echo is measured.
4. The speed of ultrasound in the material is known.
5. Distance = speed $\times$ time,
6. **using half the time (or half the distance)**, because the pulse travels there and back.

Higher frequency means shorter wavelength, so less diffraction and finer detail (resolution), and smaller cracks can be detected. Shorter pulses give better depth resolution. The echo must return before the next pulse is sent, so a longer pulse interval allows a greater range. Most of the ultrasound is reflected by the first large boundary (crack or air), so deeper features may not be detected.

8. Particle nature of light and electrons

$E = hf = hc/\lambda$	1 eV = 1.60×10^{-19} J. Number of photons per second = power $\div$ energy per photon. Shorter λ means more energy per photon, so fewer photons for the same power.
$hf = \Phi + \frac{1}{2}mv_{\max}^2$	Work function Φ : “the minimum energy required to release an electron from the surface of the metal”. Threshold frequency $f_0 = \Phi/h$.
$E_{k,\max}$ - f graph	Gradient = h ; x-intercept = f_0 ; y-intercept = $-\Phi$. Plotting E against $1/\lambda$ gives gradient hc .
$\lambda = h/p = h/mv$	de Broglie wavelength. An electron accelerated through V : $eV = \frac{1}{2}mv^2$, so $v = \sqrt{(2eV/m)}$ (use $m_e = 9.11 \times 10^{-31}$ kg).

Photoelectric effect: what the evidence shows

Observation	(1) There is a threshold frequency below which no electrons are released, however intense the light. (2) Emission is instantaneous. (3) $E_{k,\max}$ depends on frequency, not intensity. (4) Intensity only changes the number of electrons per second (so the current).
Wave model predicts	Energy is spread across the wavefront and builds up over time, so there would be a delay; any frequency would eventually release electrons; brighter light (larger amplitude) would give faster electrons.
Photon model explains	Light arrives as photons (quanta), each with $E = hf$. One photon interacts with one electron and gives all its energy. Part of the energy releases the electron ($\geq \Phi$), and the rest becomes KE, so $KE_{\max} <$ photon energy. If $hf < \Phi$ no electron escapes. More intense light means more photons per second, so more electrons per second and a larger current, but not more KE each.
Why KE varies (up to a maximum)	Electrons below the surface need more energy to escape, so they leave with less than the maximum KE.
Two-metal / two-source questions	Compare photon energy with each work function: if it lies between them, one metal emits and the other does not, which bounds the unknown work function (Jun 2021). A reversed-polarity gauze gives zero current because copper’s Φ exceeds the photon energy (Oct 2023).

Energy levels and line spectra

- **Emission (discharge tube, sodium lamp):** a current (or heating) gives atoms energy; electrons move to higher energy levels (excitation); they fall back to lower levels, emitting a photon with $hf = E_{\text{upper}} - E_{\text{lower}}$. Energy levels are **discrete**, so only certain transitions and therefore only certain frequencies (lines) occur. Different elements have different levels, so different spectra; air is a mixture of gases, so it gives many lines.

- **Absorption:** a photon is absorbed only if its energy **exactly** equals a level difference. A photon cannot give part of its energy, so a 12.3 eV photon cannot excite hydrogen from -13.6 eV to -1.5 eV (needs 12.1 eV). A colliding **electron** can give part of its KE (keeping 0.2 eV).
- **Counting lines:** from level 3 the possible downward transitions are $3 \rightarrow 2$, $3 \rightarrow 1$, $2 \rightarrow 1$, so 3 lines. The largest energy gap gives the highest frequency and shortest wavelength.
- Use “electrons” (not “atoms”) moving between levels; “excited/de-excited” is accepted.

Wave nature of electrons

- **Electron diffraction through graphite:** rings show diffraction/interference, a wave property, so electrons can behave as waves. Diffraction is significant because the de Broglie wavelength of the electrons is similar to the spacing between atoms. Graphite’s atoms are in a regular, ordered arrangement, so it acts as a grating.
- **Higher accelerating p.d.:** greater v and p , so a shorter λ , so less diffraction and smaller rings.
- **Everyday objects:** λ is far too small to observe. A car is not a single particle, and de Broglie has only been verified for microscopic particles (Oct 2019).
- Photon momentum $p = h/\lambda$ (MCQ).

EXAMPLE (JAN 2026). Electron through 4800 V: $W = 1.6 \times 10^{-19} \times 4800 = 7.68 \times 10^{-16}$ J; $v = \sqrt{(2 \times 7.68 \times 10^{-16} / 9.11 \times 10^{-31})} = 4.11 \times 10^7$ m s $^{-1}$; $\lambda = 6.63 \times 10^{-34} / (9.11 \times 10^{-31} \times 4.11 \times 10^7) = 1.8 \times 10^{-11}$ m.

9. Six-mark model answers

Each answer below contains six indicative points in order, worded as the schemes word them.

A. PHOTOELECTRIC EFFECT AND THE WAVE MODEL. JUN 2019, JAN 2022, JUN 2021, OCT 2023, JAN 2025.

1. Photon energy is proportional to frequency ($E = hf$), and there is a one-to-one interaction between a photon and an electron.
2. A minimum (threshold) frequency is needed, because the photon energy must be at least the work function. In the wave model any frequency would eventually release electrons.
3. Release of electrons is instantaneous. In the wave model energy would build up over time, so there would be a delay.
4. The (maximum) kinetic energy of the released electrons depends on frequency: $hf = \Phi + KE_{\max}$.
5. In the wave model a more intense (larger-amplitude) wave would give electrons more KE.
6. Intensity only affects the number of photons, and so the number of electrons released per second (the current).

Gauze/magnesium version (Oct 2023): UV photons cause photoelectrons to be emitted from the magnesium because the photon energy exceeds its work function; the electrons are attracted to the positive gauze, giving a current; reversed, there is no current because copper's work function is larger than the photon energy.

B. SENSOR IN A POTENTIAL DIVIDER. THERMISTOR: OCT 2019, OCT 2025; LDR POWER: JUN 2023; LDR AND INTERNAL RESISTANCE: JAN 2025.

1. As the temperature increases, electrons in the thermistor gain energy,
2. so more conduction electrons are released,
3. so the resistance of the thermistor decreases.
4. The thermistor and fixed resistor are in series, so they share the supply p.d. (a potential divider).
5. The thermistor becomes a smaller proportion of the total resistance (the current increases, so the p.d. across the fixed resistor increases),
6. so the p.d. across the thermistor decreases. Connect the switching circuit across whichever component gives the change needed.

LDR, power version: light increases, so the LDR resistance decreases (more conduction electrons), so the circuit resistance decreases and the current increases; the e.m.f. is the same, so the power $P = \varepsilon I$ increases. **Internal resistance version:** more photons on the LDR, so more conduction electrons and lower resistance; r is constant, so the larger current increases the p.d. across r , so the terminal p.d. (voltmeter reading) decreases.

C. LINE SPECTRA FROM A DISCHARGE TUBE. OCT 2021, JUN 2024; PARTS IN MANY PAPERS.

1. The p.d. causes a current in the gas, so energy is transferred to the gas atoms.
2. Electrons gain energy and move up to higher energy levels (are excited).
3. The excited electrons fall back down to lower levels, emitting a photon.
4. The photon energy equals the difference between the levels, with $E = hf$.
5. The energy levels are discrete, so only certain energy differences are possible.
6. So only certain frequencies/wavelengths are emitted ($v = f\lambda$), giving a line spectrum.

D. INTERFERENCE OF WAVES FROM TWO GAPS OR TWO SOURCES. JAN 2023, JAN 2024; HEADPHONES OCT 2020; BEATS OCT 2022.

1. Waves diffract (spread out) as they pass through each gap.
2. The waves from the two gaps overlap and superpose (interfere).
3. At X (the loud point) the path difference is zero, so the waves are in phase.
4. So constructive interference gives a maximum amplitude (loud).
5. Moving away from X, the path difference changes.
6. Where the path difference is $\lambda/2$ (antiphase), destructive interference gives a minimum (no sound).

Noise-cancelling headphones: the anti-noise must diffract to reach the far ear; it cancels where it meets the noise in antiphase (destructive) and is louder where they are in phase; this depends on the ear separation compared with λ . **Two slightly different frequencies (beats):** the waves are not coherent, so the phase difference changes continuously, alternating between in phase (loud, constructive) and antiphase (quiet, destructive). Answer in phase terms, not path difference.

E. STATIONARY WAVE IN A TUBE CLOSED AT ONE END. JUN 2025; STRING VERSIONS IN MANY PAPERS.

1. The vibrating tuning fork produces sound waves in the air in the tube.
2. The sound wave reflects from the closed end.
3. The reflected wave superposes (interferes) with the incident wave.
4. Where they are in phase, constructive interference gives maximum amplitude (antinodes); where they are in antiphase, destructive interference gives zero amplitude (nodes).
5. There is a node at the closed end
6. and an antinode at the open end, so the shortest resonating length is $\lambda/4$.

F. OTHER CONTEXTS (ONE-LINE CHAINS)

- **Filament lamp vs thermistor I - V (Jan 2021):** lamp: resistance increases as V increases, because the higher temperature makes the ions vibrate more, so collisions with electrons increase. Thermistor: resistance decreases, because more conduction electrons are released, and $I = nqvA$, so current increases faster than p.d.
- **Current and p.d. rules in a series-parallel circuit (Jun 2022):** current is the rate of flow of charge and is the same at all points in series; current into a junction = current out; p.d. is energy per unit charge; p.d.s in series add to the e.m.f.; p.d. is the same across parallel branches.
- **Light in a swimming pool (Oct 2024):** light is slower in water (n larger), so ray A refracts towards the normal. Ray B meets the surface at more than the critical angle, so it is totally internally reflected. Ray C is below the critical angle, so it refracts out.
- **Ultrasound baby scan / crack detection (Jan 2020, Jan 2026):** use the six steps in the box in section 7.

10. Multiple-choice traps seen in these papers

- Resistance on an I - V curve is V/I at the point, not $1/\text{gradient}$ of the tangent.
- A single polarising filter on unpolarised light: intensity does not change as it rotates.
- Polarisation proves light is transverse; diffraction, refraction and superposition happen to all waves.
- Waves on a vibrating string forming a pattern are stationary, not progressive.
- Compressions and rarefactions both have zero particle displacement at their centres.
- Wavelength cannot be read from a displacement-time graph.
- Entering a denser medium: frequency unchanged; speed, wavelength (and transmitted amplitude) decrease.
- TIR needs higher $n \rightarrow$ lower n **and** $i > C$; at $i > C$ **all** light is reflected.
- Ultrasound distance = $\frac{1}{2}vt$; convert ms and μs first.
- Intensity from the Sun: $I(4\pi r^2)$ is constant; the 4π cancels in ratios, but the radius must be squared.
- $v \propto \sqrt{T}$: halving the length of the same string (same μ) does not change v for the same tension.
- Intensity changes the number of photoelectrons; frequency changes their KE.
- A photon must match a level difference exactly; an electron can give part of its energy.
- Photon momentum is h/λ ; electron λ uses m_e , not the proton mass.
- n in $I = nqvA$ is carriers per m^3 (not per metre, not total electrons).
- Thermistor heated: more conduction electrons **and** larger lattice vibrations; the first dominates.
- Power in the internal resistance = $I \times$ (p.d. across r) = I^2r .
- Path difference λ gives constructive interference; $\lambda/2$ or 1.5λ gives destructive.
- Increasing light intensity on an LDR in a divider: the p.d. across the LDR falls and across the fixed resistor rises.

11. Final checklist before the exam

Can you, without notes...	Can you, without notes...
Derive the series and parallel resistance formulae, naming conservation of charge and energy?	Write the polarisation definition with “one plane including the direction of travel”?
Solve a potential divider by ratio and predict all meter changes?	Explain how a stationary wave forms and find λ for a string and a closed tube?
Get ε and r from a V - I graph and draw the circuit?	Find lines per mm from $\tan \theta$ and $n\lambda = d \sin \theta$, and the maximum order?
Do $R = \rho l/A$ and $I = nqvA$ with correct areas (circle or strip)?	Decide whether TIR occurs and explain refraction with wavefronts?
Explain metal vs thermistor resistance changes with temperature?	Write the photoelectric 6-marker and the line-spectra chain from memory?
Convert eV $\leftrightarrow$ J and find photons per second from a power?	Find a de Broglie wavelength for an electron accelerated through V ?

Data: $h = 6.63 \times 10^{-34}$ J s; $c = 3.00 \times 10^8$ m s⁻¹; $e = 1.60 \times 10^{-19}$ C; $m_e = 9.11 \times 10^{-31}$ kg; 1 eV = 1.60×10^{-19} J; speed of sound in air ≈ 340 m s⁻¹. Formulae sheet: $I = \Delta Q/\Delta t$, $V = W/Q$, $R = V/I$, $P = VI$, $W = VIt$, $R = \rho l/A$, $I = nqvA$, $\varepsilon = I(R + r)$, $v = f\lambda$, $v = \sqrt{T/\mu}$, $I = P/A$, $n_1 \sin \theta_1 = n_2 \sin \theta_2$, $\sin C = 1/n$, $n\lambda = d \sin \theta$, $E = hf$, $hf = \Phi + \frac{1}{2}mv_{\max}^2$, $\lambda = h/p$.